

Numerical Study of Archimedes Propeller Configurations

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This paper presents numerical simulations of five Archimedes propeller (AP) configurations. All configurations share the same diameter and length, with the number of blades varying from 2 to 5. The Delayed Detached Eddy Simulation (DDES) approach using the Shear Stress Transport (SST) $k-\omega$ model, coupled with the Ffowcs Williams–Hawkings (FW–H) acoustic analogy in ANSYS Fluent, is employed to investigate propeller performance, aeroacoustics, and flow characteristics. Preliminary results for the AP-3 configuration indicate moderate performance based on the figure of merit (FM). Propeller thrust is primarily generated by pressure differences near the blade leading edges. This configuration also exhibits irregular noise directivity, particularly for the primary tonal noise. The helical blade shape leads to distinct tip vortex formations. The final paper will include numerical results from the remaining AP configurations, enabling comparative analysis and providing insights and recommendations for practical applications in small drones.

Nomenclature

Symbol

A_{disk}	=	propeller disk area (m^2)
c_0	=	ambient speed of sound (m/s)
C_p	=	static pressure coefficient
C_P	=	propeller power coefficient
$C_{P_{max}}$	=	maximum power coefficient
C_Q	=	propeller torque coefficient
C_T	=	propeller thrust coefficient
d	=	wall distance (m)
D	=	propeller diameter (m)
D_ω	=	cross-diffusion term
f	=	noise frequency (Hz)
f^+	=	nondimensional noise frequency
G_ω	=	generation of ω
$\tilde{G}_k$	=	generation of turbulence kinetic energy
$H(f)$	=	Heaviside function
k	=	turbulent kinetic energy
N	=	the number of the propeller blades
P_{ij}	=	compressive stress tensor
p	=	static pressure (Pa)
p'	=	acoustic pressure (Pa)
Q	=	propeller torque (N·m)
Q_{norm}	=	normalized Q-criteria
Q_{raw}	=	raw Q-criteria (m^2/s^2)
R	=	propeller radius (m)
S_{ij}	=	strain-rate tensor
T	=	propeller thrust (N)

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T_{ij}	=	Lighthill stress tensor
u_i	=	velocity component (m/s)
U_{tip}	=	propeller tip speed (m/s)
v	=	moving velocity of the data surface (m/s)
y^+	=	dimensionless wall distance
Y_k	=	dissipation of k
Y_ω	=	dissipation of ω
ν	=	molecular viscosity
ν_t	=	turbulent eddy viscosity
ρ	=	air density (kg/m ³)
χ	=	propeller tradeoff index
ω	=	specific dissipation rate
ω_z	=	z-vorticity (1/s)
Ω	=	rotational speed (RPM)
Γ_k	=	effective diffusivity of k
Γ_ω	=	effective diffusivity of ω
$\delta(f)$	=	Dirac delta function
Δt	=	time step

Abbreviation

AP	Archimedes propeller
ASWT	Archimedes Spiral Wind Turbine
BPF	blade passage frequency (Hz)
CFD	computational fluid dynamics
dB	decibel
DDES	Delayed Detached Eddy Simulation
FFT	fast Fourier transform
FM	figure of merit
$freq$	maximum frequency of interest (Hz)
FW-H	Ffowcs-Williams and Hawkings
LES	large eddy simulation
OASPL	overall sound pressure level (dB)
RANS	Reynolds Averaged Navier–Stokes
RPM	revolution per minute
SA	Spalart-Allmaras
SPL	sound pressure level (dB)
SST	Shear Stress Transport
TKE	turbulent kinetic energy
UAM	urban air mobility
UMD	University of Maryland

I. Introduction

Recent advances in electric propulsion, lightweight materials, and autonomous flight technologies have accelerated the development of propeller-driven systems for both urban air mobility (UAM) and renewable energy applications. As these systems are increasingly deployed in densely populated and noise-sensitive environments, minimizing their acoustic footprint while maintaining aerodynamic performance has become a critical design challenge. Beyond UAM, noise generated by wind turbines continues to pose significant environmental concerns, often disrupting nearby ecosystems. Among the various turbine concepts explored, the Archimedes Spiral Wind Turbine (ASWT) has attracted growing

attention. Although still in its early stages of development, the Archimedes Spiral configuration has been applied both to wind turbine designs, like the Liam F1 turbine [1], and to a small drone prototype developed at the University of Maryland (UMD).

Table 1 summarizes several previous studies that examined the ASWT, outlining their analysis methods, comparison conditions, and reported maximum power coefficient ($C_{P_{max}}$) values. While these investigations have primarily evaluated the ASWT as a wind turbine [2], the present study adopts a different perspective—examining its operation as a propeller, hence the name Archimedes Propeller (AP). Interestingly, researchers and students at UMD developed a drone prototype, “CrimsonSpin,” which utilized similar spiral-shaped propellers and received recognition at the Vertical Flight Society’s 2022 Transformative Vertical Flight Conference.

Table 1 Comparison of ASWT studies

Previous Work	Method	Focus	TSR	Condition	$C_{P_{max}}$
Jang and Kim [2]	$k - \omega$ in RANS & field test	Performance validation	2.19	Same propeller	0.293
Kamal and Nawar [3]	$k - \omega$ in RANS & experiment (open wind tunnel)	Blade angle & Blade thickness	1.5–2.0	Varied blade angles	0.243
Song and Huan [4]	$k - \omega$ in RANS	Blade angle	1.5	Five blade-angle configurations	0.234
Hamid and Abd [5]	$k - \omega$ in RANS	Opening blade angle	2.5	Opening angle varied from 40° to 90°	0.268
Safdari and Kim [6]	LES & structural evaluation	Aerodynamic characteristics and structural assessment	–	Same propeller	0.25
Nawar and Hameed [7]	$k - \omega$ in RANS & experiment	Effect of blade design (fixed vs variable)	Fixed: 1.57, Variable: 1.96	Fixed vs variable design angle	Variable: 0.226, Fixed: 0.207

TSR: tip speed ratio; RANS: Reynolds Averaged Navier–Stokes; LES: large eddy simulation

Unlike previous studies in Table 1, this work focuses on evaluating Archimedes Spiral configurations in their role as propellers. The Delayed Detached Eddy Simulation (DDES) approach was employed based on prior comparisons by the third author between experimental results and predictions from various flow solvers. Among the propulsion-related challenges, propeller noise is one of the most pressing issues for UAM drones [8]. To mitigate these acoustic challenges, various noise-reduction strategies have been proposed, including serrated trailing edges [9] and active flow control techniques [10]. However, these methods often provide limited noise reduction or incur additional costs. Recently, there has been growing interest in unconventional propeller geometries that may modify flow behavior and noise generation mechanisms, such as toroidal propellers [11–20]. The AP is another such configuration, characterized by its helical blade geometry, smooth curvature, and increased thickness along the propeller shaft axis. This geometry promotes gradual pressure gradients and weaker tip vortices, potentially offering inherent noise-reduction benefits. Despite these promising characteristics, the aeroacoustics of the AP remains largely unexplored. Understanding how this unique geometry influences both aerodynamic performance and acoustic emissions is essential for its application in UAM and renewable energy systems.

To address this knowledge gap, this study investigates the aerodynamic and aeroacoustic performance of five AP configurations in hover using numerical simulations. Commercial computational fluid dynamics (CFD) software was employed to evaluate propellers with two to five blades, aiming to identify the configuration that achieves the highest score according to a propeller tradeoff index [21]. The computational domain was modeled as a cylindrical enclosure consisting of a rotating domain, a stationary domain, and a refinement zone, all concentric with the propeller, which has a diameter of 0.142 m. Data obtained from the CFD analysis include propeller performance, acoustic characteristics, and flowfield patterns. The objective of this study is to quantify thrust generation and acoustic behavior while visualizing the flow mechanisms associated with the AP’s distinctive geometry. The results presented here establish a foundation for future studies that will further explore and optimize this design.

II. Numerical Methods

A. Propeller Geometry

To numerically investigate the aerodynamic and aeroacoustic characteristics of the AP configuration, four geometrical variants were modeled using SOLIDWORKS (Version 2025), designated as AP-2 (Fig. 1a), AP-3 (Fig. 1b), AP-4 (Fig. 1c), and AP-5 (Fig. 1d). The numerical identifier corresponds to the number of blades incorporated in each configuration, ranging from 2 to 5. All propeller models were designed with an identical diameter ($D = 0.142$ m, largest), overall length (0.127 m), and pitch angle to ensure a consistent basis for comparison. The blades adopt a flat airfoil profile with smoothly rounded leading and tips to emulate the continuous helical surface characteristic of the Archimedes-type geometry.

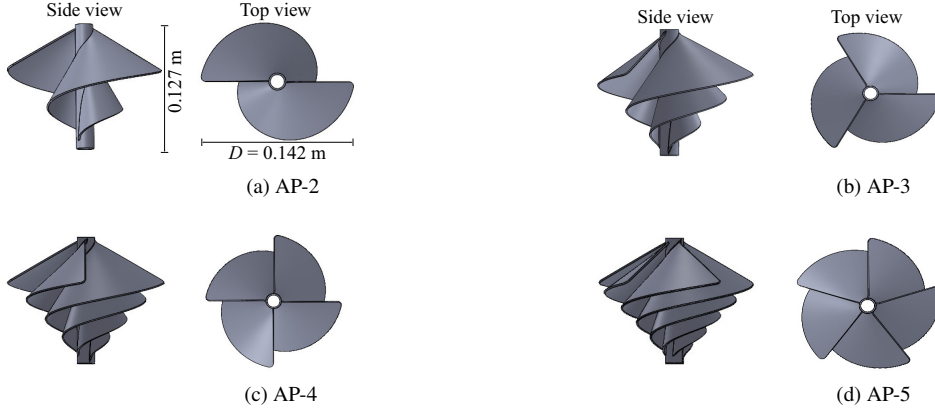


Fig. 1 Images of Archimedes propeller configurations.

B. Flow modeling

The DDES approach was chosen to simulate the flow, with the Shear Stress Transport (SST) $k - \omega$ model employed for turbulence modeling. The governing equations for the incompressible flow [22] are:

$$\frac{\partial u_i}{\partial t} + \frac{\partial(u_i u_j)}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left[(\nu + \nu_t) \frac{\partial u_i}{\partial x_j} \right] \quad (1)$$

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (2)$$

The turbulent eddy viscosity ν_t is modified via DDES switching:

$$\nu_t = \rho \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (3)$$

where the relevant functions and terms are defined as

$$SF_2 = \sqrt{2S_{ij}S_{ij}} \quad \text{and} \quad S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (4)$$

Here, ν is the molecular viscosity, and S_{ij} is the strain-rate tensor. Default values of the constants are used, i.e., $a_1 = 0.31$, $C_{DES} \text{ (Inner)} = 0.78$, $C_{DES} \text{ (Outer)} = 0.61$, $C_{d1} = 20$, $\alpha_\infty^* = 1$, $\alpha_\infty = 0.52$, $\beta_\infty^* = 1$, $\beta_i \text{ (Inner)} = 0.075$, and $\beta_i \text{ (Outer)} = 0.0828$.

DDES combines the strengths of RANS and LES methods: the boundary layer is handled by RANS, while regions with significant flow separation are resolved using LES. Consequently, near-wall turbulence is approximated rather than fully resolved, allowing for a coarser mesh near solid surfaces and reducing the demand for fine-grid resolution.

The DDES model, detailed by Spalart et al. [23], is an adaptation of the original DES97 model through the introduction of a shielding function f_d that prevents premature switching to LES mode within the boundary layer. This limiter is defined in Eq. 5.

$$f_d = 1 - \tanh([8r_d]^3) \quad (5)$$

where

$$r_d = \frac{\nu_t + \nu}{\omega d^2 \sqrt{S_{ij} S_{ij}}} \quad (6)$$

Here, d is the wall distance. While Eqs. 5 and 6 were originally developed for the Spalart-Allmaras (SA) DES model, the DDES implementation introduces a modified DES length scale $\tilde{d}$, defined in Eq. 7.

$$\tilde{d} = d - f_d \max(0, d - C_{DES}\Delta) \quad (7)$$

In this formulation, $\tilde{d}$ depends on the grid size $\Delta = \max(\Delta x, \Delta y, \Delta z)$ and the eddy-viscosity field through the f_d limiter.

The SST $k - \omega$ turbulence model, originally developed by Menter [24], governs turbulence in the boundary layer and is defined by the transport equations for turbulent kinetic energy, k , and specific dissipation rate, ω , as presented in Eqs. 8 and 9.

$$\frac{\partial}{\partial t}(\rho k) + \frac{\partial}{\partial x_i}(\rho k u_i) = \frac{\partial}{\partial x_i}(\Gamma_k \frac{\partial k}{\partial x_j}) + \tilde{G}_k - Y_k + S_k \quad (8)$$

$$\frac{\partial}{\partial t}(\rho \omega) + \frac{\partial}{\partial x_i}(\rho \omega u_i) = \frac{\partial}{\partial x_i}(\Gamma_\omega \frac{\partial \omega}{\partial x_j}) + G_\omega - Y_\omega + D_\omega + S_\omega \quad (9)$$

Here, ρ is the air density, u_i is the velocity component, Γ_k and Γ_ω represent the effective diffusivity of k and ω , respectively. Y_k and Y_ω denote the dissipation of k and ω due to turbulence, while S_k and S_ω represent user-defined terms. $\tilde{G}_k$ represents the generation of turbulence kinetic energy due to mean velocity gradients, G_ω is the generation of ω , and D_ω is the cross-diffusion term.

C. Acoustics computation

Acoustic predictions were conducted using the Ffowcs-Williams and Hawkings (FW-H) acoustic analogy, formulated as:

$$\frac{1}{c_0^2} \frac{\partial^2 p'}{\partial t^2} - \nabla^2 p' = \frac{\partial^2}{\partial x \partial y} [T_{ij} H(f)] - \frac{\partial}{\partial x_i} [P_{ij} n_j + \rho u_i (u_n - v_n)] \delta(f) + \frac{\partial}{\partial t} [\rho_0 v_n + \rho (u_n - v_n)] \delta(f) \quad (10)$$

In this equation, p' represents the acoustic pressure, c_0 is the ambient speed of sound, with the subscript "0" denotes ambient condition, and v is the moving velocity of the data surface. The subscript n denotes the local normal term of the data surface. T_{ij} and P_{ij} correspond to the Lighthill stress tensor and the compressive stress tensor, respectively, while $H(f)$ and $\delta(f)$ are the Heaviside and Dirac delta functions. This formulation, as implemented in Zhi et al. [25], is consistent with the original approach of Williams and Hawkings [26]. The FW-H solution includes contributions from quadrupole sources (e.g., turbulence-related noise), dipole sources (e.g., surface force-related noise), and monopole sources (e.g., mass fluctuation-related noise).

All propeller surfaces were treated as FW-H integration surfaces for sound generation. Specifically, using DDES data from the final 20 propeller revolutions, transient acoustic fields were computed via the built-in FW-H acoustic model in ANSYS Fluent. The resulting acoustic pressure time series at selected receiver locations were post-processed using fast Fourier transform (FFT) to obtain sound pressure spectra.

In the transient simulations, the time step (Δt) was determined using Eq. 11 to ensure sufficient temporal resolution for the maximum frequency of interest. A value of set to $\Delta t = 8.33 \times 10^{-5}$ seconds was used, and the total simulation time was one seconds.

$$\Delta t \leq \frac{1}{2 * freq} \quad (11)$$

where $freq$ denotes the maximum frequency of interest.

III. Computational setup

A. Domain

Numerical simulations were carried out in ANSYS Fluent (Version 21.2), a commercial CFD solver. A velocity inlet boundary condition with a magnitude of 0.1 m/s was applied at the inlet, and a pressure outlet was used to define the end of the computational domain, as illustrated in Fig. 2. To model the rotation of the hybrid propeller, a small rotating domain with a diameter of $1.1D$ and a depth of $1D$ was included within the stationary domain in meshing motion. The stationary domain is cylindrical with dimensions of $12D$ by $30D$ (diameter by length). The height from the center of the propeller to the inlet is $10D$ and to the outlet is $20D$. To capture the propeller wake more accurately, a refinement region is introduced: it extends from $1D$ above the propeller center to $8D$ below it and maintains a constant diameter of $2D$.

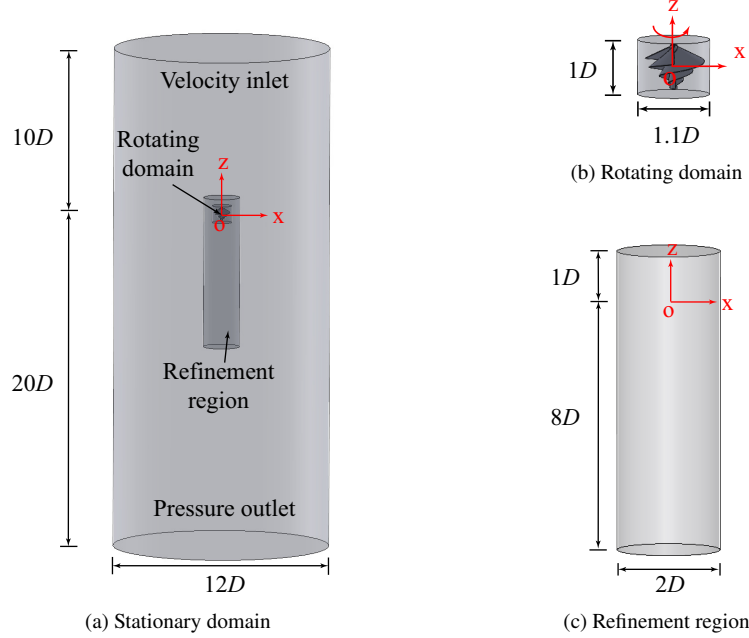


Fig. 2 Presentation of simulation domains in ANSYS Fluent.

B. Grid

The computational domain was discretized utilizing an unstructured, polyhedral mesh generated with the meshing module in ANSYS Fluent. AP-3 configuration is used for presentation. A refinement region was included around the rotating domain to improve resolution in the propeller wake, as shown in Figure 3a. Figure 3b presents the grid near the propeller model in the xz -plane. Inflation layers, illustrated in Fig. 3c, were applied along the toroidal propeller surfaces to enhance boundary layer resolution—a critical factor for accurately capturing near-wall flow characteristics—alongside appropriate face sizing. This setup included ten layers with a growth (transition) ratio of 0.10 and an initial layer height of 0.01 mm, resulting in a y^+ value below 1 for the first layer above the propeller surfaces. The final mesh for the AP-3 configuration, which represents the completed portion of the study presented in the abstract, consisted of approximately 15.7 million cells.

C. Solver

The pressure-based coupled algorithm was employed to resolve the pressure–velocity coupling, enhancing numerical stability and accelerating convergence in the transient flow simulation. Temporal discretization was carried out using a bounded second-order implicit scheme, which provides both stability and second-order accuracy for unsteady simulations.

The spatial discretization schemes were selected to ensure a balance between accuracy and numerical stability. The gradient was computed using the least squares cell-based method, which offers robustness on unstructured meshes. The

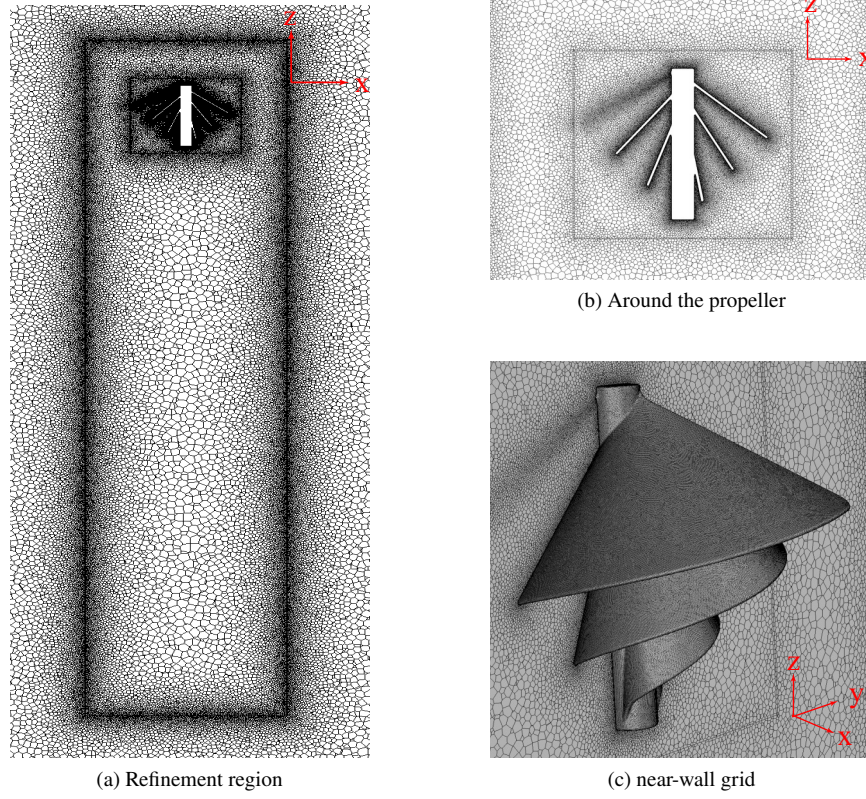


Fig. 3 Grid presentation of the AP-3 configuration in ANSYS Fluent.

pressure term was discretized using a second-order scheme, while the momentum equations were handled with the bounded central differencing scheme, suitable for LES-like behavior in DDES.

The turbulence quantities, including the turbulent kinetic energy (TKE, k) and specific dissipation rate (ω), were discretized using the second-order upwind scheme, preserving accuracy in regions with steep gradients. The residual convergence criterion was set to 10^{-6} for all equations, and propeller thrust and torque were monitored during the simulations.

D. Data processing

1. Propeller aerodynamics

Propeller thrust (T) and torque (Q) measurements were nondimensionalized using Eqs. 12 and 13 [27], where A_{disk} is the propeller disk area, U_{tip} is the tip speed, R is the propeller radius.

$$C_T = \frac{T}{\rho A_{disk} U_{tip}^2} \quad (12)$$

$$C_Q = \frac{Q}{\rho A_{disk} U_{tip}^2 R} \quad (13)$$

Hover figure of merit (FM) is a measure of propeller performance and is defined as the ratio between the ideal power required to hover and the actual power required [28],

$$FM = \frac{C_T^{3/2}}{\sqrt{2} C_P} = \frac{C_T^{3/2}}{\sqrt{2} C_Q} < 1 \quad (14)$$

For clarification, using the above nondimensionalization method, C_P and C_Q are numerically equivalent.

2. Propeller aeroacoustics

To process the acoustic results from the computations, the methodology is described in detail by Chen et al. [8]. The key steps for aeroacoustic data processing are summarized as follows:

- **Tonal noise:** Frequencies obtained from the fast Fourier transform (FFT) are normalized by the blade passing frequency (BPF), as defined in Eq. 11. Tonal components are identified as peaks occurring at integer values of the normalized frequency, f^+ . In this study, the analysis focuses on the primary tonal noise corresponding to $f^+ = 1$.
- **Broadband noise:** Noise peaks are removed using a built-in function in MATLAB 2021.
- **Overall sound pressure level (OASPL):** The noise levels are integrated over the entire frequency range.

$$f^+ = \frac{f}{\text{BPF}} \quad (15)$$

The BPF is defined in Eq. 16:

$$\text{BPF} = \frac{N \cdot \Omega}{60} \quad (16)$$

where N is the number of blades on the Archimedes propellers, and Ω is the rotational speed in revolutions per minute (RPM).

IV. Initial Results

In this section, simulation results for the AP-3 configuration operating at 3,000 RPM are presented. The analysis initially focuses on propeller performance, aeroacoustic characteristics, and tip vortex structures.

A. Propeller performance

Figure 4a presents thrust coefficient (C_T), torque coefficient (C_Q), and FM for the AP-3 configuration. A FM of 0.378 indicates that this configuration has moderate propeller performance. To understand it, static pressure coefficient contours on the propeller surfaces are shown in Fig. 4b, with the coefficient defined by Eq. 17.

$$C_p = \frac{p}{\frac{1}{2}\rho U_{tip}^2} \quad (17)$$

where p denotes the static gauge pressure. The pressure contours reveal that the thrust generation primarily originates from the pressure difference near the leading edges of the blades. In contrast, the remaining blade surfaces exhibit nearly uniform pressure close to zero, suggesting that these regions contribute minimally to thrust. Consequently, trimming or tapering these portions could potentially reduce the torque induced by skin friction as the propeller rotates in air.

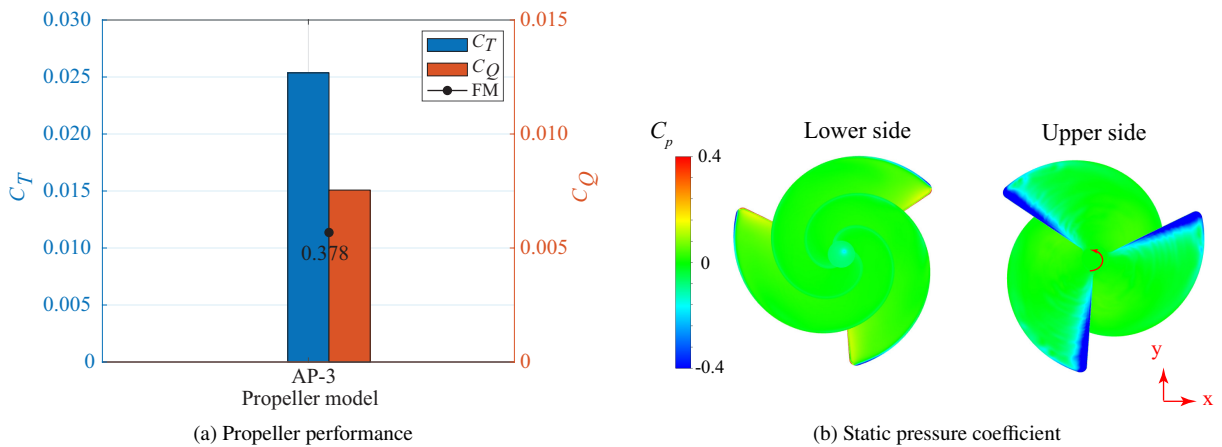


Fig. 4 Performance results of the AP-3 configuration.

B. Propeller aeroacoustics

To examine the aeroacoustic characteristics of the AP-3 propeller, 13 receivers were placed in the xz -plane at 15° intervals, each located 0.3505 m from the origin, as illustrated in Fig. 5a. The directivity pattern of the primary tonal noise at the blade-passing frequency ($f^+ = 1$), shown in Fig. 5b, exhibits an irregular distribution. Similar irregularities are observed for the broadband noise (Fig. 5c) and the OASPL (Fig. 5d). The smallest primary tonal noise, approximately 32 dB, occurs at 45° . In contrast, both broadband noise and OASPL display a more uniform directivity pattern. The results also suggest that the direction of rotation influences the overall noise directivity.

The sound pressure levels (SPL) at receiver locations R1 and R4 are plotted in Fig. 5e and Fig. 5f, respectively. The most notable difference between these two locations is the amplitude of the primary tonal noise. Minor variations are also observed in the SPL spectra at frequencies below $f^+ < 1$.

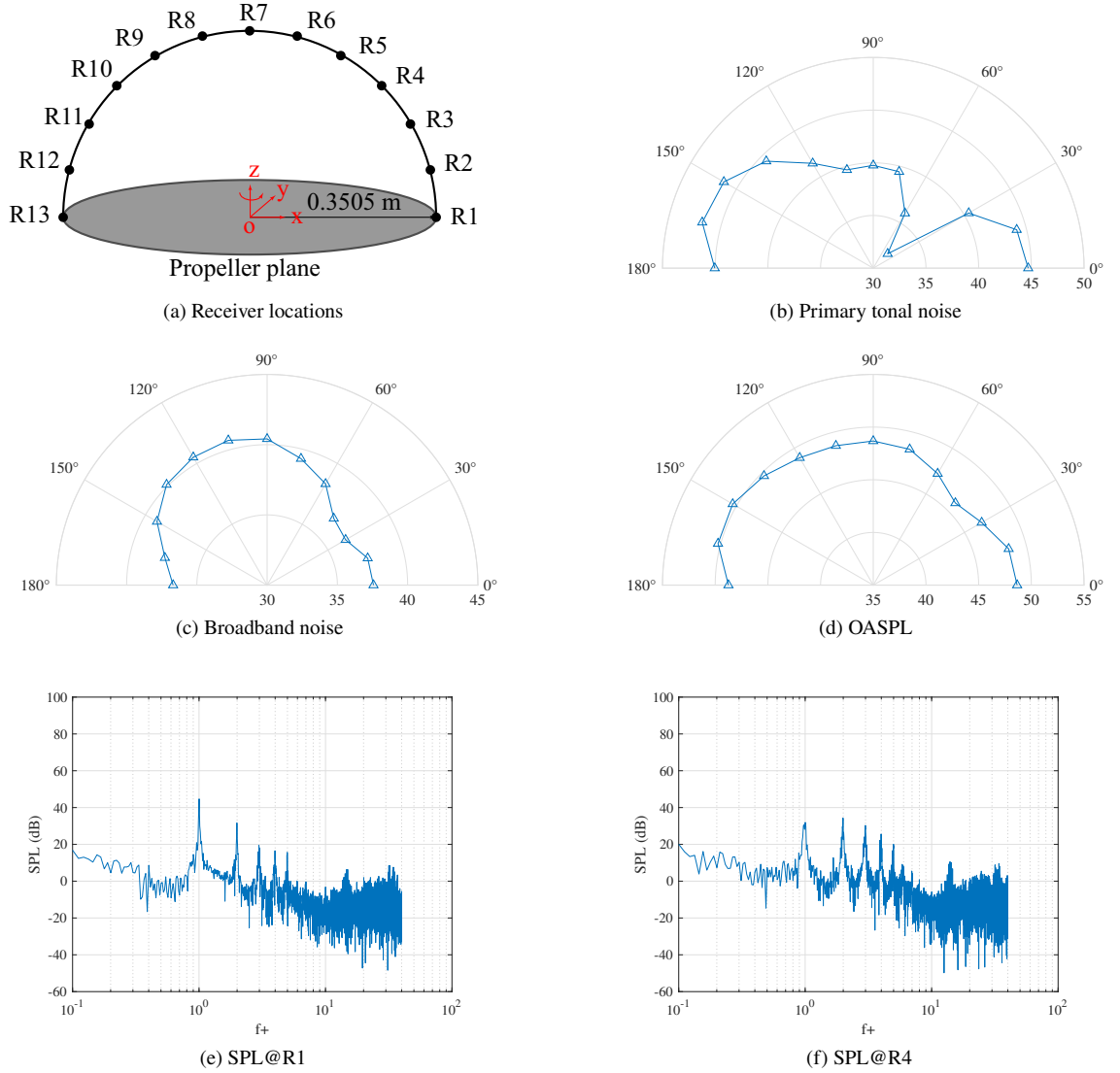


Fig. 5 Aeroacoustics results of the AP-3 configuration.

C. Tip vortices

To analyze the tip vortices, the Q-criterion—a widely used method for identifying vortex cores in turbulent flows—is employed in its nondimensional form:

$$Q_{norm} = \frac{Q_{raw}}{U_{tip}^2}. \quad (18)$$

Here, Q_{raw} is obtained directly from ANSYS Fluent. The Q -iso-surfaces ($Q_{norm} = 0.8$) are further colored by the normalized z -vorticity ($\frac{\omega_z D}{U_{tip}}$), representing the local rotational or swirling motion of the flow, as shown in Fig. 6a. Owing to the spiral blade geometry, multiple tip vortices are generated along the blade tips. After shedding from the tips, these vortices convect downward in a helical pattern due to the blade rotation and helical angle, with their trajectories gradually contracting in radius. A cross-sectional view on the yz plane is presented in Fig. 6b, revealing that strong tip vortices are absent near the bottom portion of the propeller.

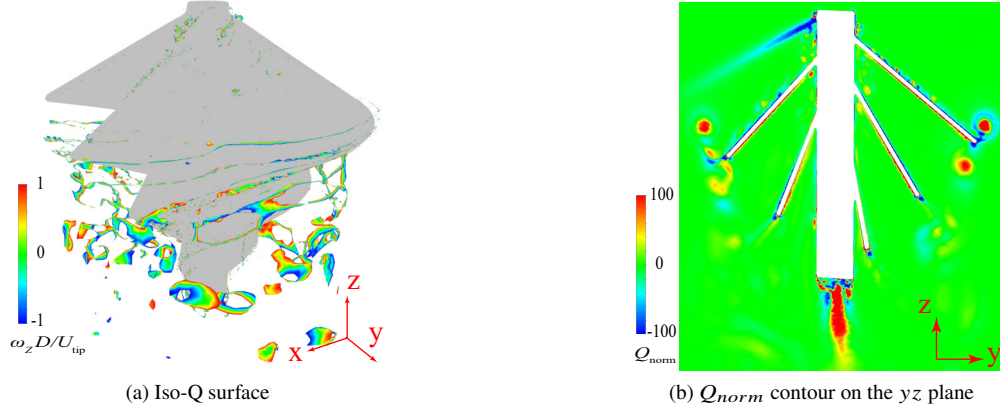


Fig. 6 Tip vortex results of the AP-3 configuration.

V. Final Paper

The final version of the paper will include the following:

- 1) Comprehensive analysis of propeller performance, aeroacoustics, and tip vortex behavior by incorporating results from all four AP configurations.
- 2) Quantitative characterization of tip vortex trajectories for each AP configuration.
- 3) Visualization of flow features, including streamlines on the yz -plane, turbulence intensity, and dp'/dt distributions for all AP configurations.
- 4) Evaluation of propeller performance and aeroacoustics using a propeller tradeoff index (χ) to facilitate comparative assessment among different configurations.

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